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Lessons

The Order of Operations

The expression $3 + 4 \times 5$ contains two operations: addition and multiplication. Remember that multiplication can be represented by a multiplication dot.

What is the value of the expression?

If you add first, it equals 35.	$3 + 4 \times 5$	If you multiply first, it equals 23.	$3 + 4 \times 5$
	$= 7 \times 5$		$= 3 + 20$
	$= 35$		$= 23$

Both 35 and 23 can't be right. To make sure people agree on the value of an expression, mathematicians developed rules called the **order of operations.** The following example demonstrates how to use the order of operations.

Example 1

Hide

Find the value of the expression $(3 - 7)^2 + 3(4) - \sqrt{2 \times 32}$.

Step 1

Perform operations in grouping symbols: $(3 - 7)^2 + 3(4) - \sqrt{2 \times 32}$

parentheses, brackets, fraction bars, and radicals. $= (-4)^2 + 3(4) - \sqrt{64}$

Step 2

Evaluate powers (exponents) and roots. $= (-4)^2 + 3(4) - \sqrt{64}$

$= 16 + 3(4) - 8$

Step 3

Multiply and divide, working from left to right. $= 16 + 3(4) - 8$

$= 16 + 12 - 8$

Step 4

Add and subtract, working from left to right. $= 16 + 12 - 8$

$= 28 - 8$

$= 20$

The value of the expression is **20.**

Good test takers use memory tricks to help them remember rules. To remember the order of operations, many students use the sentence "Please Excuse My Dear Aunt Sally," or PEMDAS. The first letter of each word is an order clue.

P represents **P**arentheses and other grouping symbols.

E represents **E**xponents and roots.

M and **D** represent **M**ultiplication and **D**ivision.

A and **S** represent **A**ddition and **S**ubtraction.

The next example uses a fraction bar as a grouping symbol. A fraction bar expresses division. Find the value of the numerator and the denominator separately. Then divide.

Example 2

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Find the value of the expression $\frac{(6+1)^2-4}{6-1}$.

Step 1

Find the value of the numerator and the denominator separately. Notice how the order of operations is followed in the numerator. $\frac{(6+1)^2-4}{6-1} = \frac{(7)^2-4}{6-1} = \frac{49-4}{6-1} = \frac{45}{5}$

Step 2

Divide. $\frac{45}{5} = 9$

The value of the expression is **9.**

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